

System Engineering Process

CSE3001 Software Engineering Module -I The System Engineering Process

Systems Engineering

- Designing, implementing, deploying and operating systems which include **hardware, software and people**
- It is a mix of HR, project management, business, rational decomposition, trade studies, requirements traceability, integration, testing, verification and validation, operations, and end of life cycle disposal of systems
- Standardizes the ***flow-down and traceability*** of specifications for complex products from customer requirements through production, operation, and disposal

Systems Engineering

- Systems Engineering is an ***interdisciplinary approach*** and means to enable the realization of successful systems.
- It focuses on defining ***customer needs*** and required functionality early in the development cycle, documenting requirements, then proceeding with design synthesis and system validation
- Systems Engineering considers both the business and the technical needs of all customers with the goal of ***providing a quality product*** that meets the user needs

Systems Engineering & Component Engineering

- Science
 - Determines what Is
- Component Engineering
 - Determines what Can Be
- Systems Engineering
 - Determines what Should Be